
nginx_passwd

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Gene C

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NGINX-PASSWD

1.1 Overview

Basic Auth Password Manager.

Manages basic auth password files. This replaces the functionality provided by `htpasswd` from Apache. It also provides legacy and modern hash functions, including the `argon2` and `pbkdf2_sha512`.

- All git tags are signed with arch@sapience.com key which is available via WKD or download from <https://www.sapience.com/tech>. Add the key to your package builder gpg keyring. The key is included in the Arch package and the `source=` line with `?signed` at the end can be used to verify the git tag. You can also manually verify the signature

1.2 New Or Interesting

Version 3.3.0

- Source code reorg
- Switch python packager from hatch to uv
- Confirm all working with python 3.14.2

Version 3.1.0

- Remove `os.getlogin()` in tests (ENOTTY reported by @ccharabaruk on AUR package).
- Small tidy ups

Version 3.0.0

- Reduce our reliance on `passlib` (it is unmaintained).
 - `passlib bcrypt` is broken
 - there is a fork which works
- Algo changes - limit use of `passlib` only to legacy hashes
 - `bcrypt` : now uses `python-bcrypt` directly
 - `argon2` : now uses `python-argon2-cffi` (which calls C library)
 - `pbkdf2_sha512` : now uses `python-cryptography`
 - `pbkdf2_sha256` : now uses `python-cryptography`
 - `md5_crypt` : legacy algo using `passlib`
 - `apr_md5_crypt` : legacy algo using `passlib`

Older

- PEP-8, PEP-257, PEP-484 and PEP 561
- Refactor code
- Add pytest
- add python-bcrypt dependency to Arch PKBUILD
- For simplicity, make argon2 required instead of optional

GETTING STARTED

2.1 nginx_passwd application

2.1.1 Usage

To add or modify a user and write the resulting password file:

```
nginx-passwd -f <password_file> <user>
```

If file is not specified, then the result is written to stdout.

The supported algorithms are:

```
* Modern : argon2, pbkdf2_sha512, pbkdf2_sha256
* Active  : sha512, sha256, bcrypt
* Deprecated: md5, apr_md5/apr1
```

with *sha256* being the default.

Note: At some point in future we will change the default algo to the more modern *argon2*.

Older and now deprecated algorithms (*md5*, *apr_md5*) are still supported but should be replaced by one of the active ones.

Note: *apr_md5*, also known as *apr1*, is the ancient Apache variant of md5.

Note: If the password is not provided with *-p* option and is nowhere is found to read it from, then it will be an empty string.

2.1.2 Options

The options are given below and *nginx-passwd -h* provides a help summary.

Positional Argument:

- username
required argument.
- *(-h, -help)*
show help message and exit
- *(-f, -passwd_file) <password_file>*
Write to this Password file

- *(-a, -algo)* <algorithm>

Default is now argon2 (older versions used sha256). Can be one of:

```
* Modern : argon2, pbkdf2_sha512, pbkdf2_sha256
* Active  : sha512, sha256, bcrypt
* Deprecated: md5, md5_apr1
```

- *(-p, -passwd)* <password>

Password as an option. Without this it will be read from stdin.

- *(-D, -delete)*

Delete this user from the password file.

- *(-v, -verify)*

Checks that the provided password matches that in the password file

3.1 Installation

Available on * [Github](#) * [Archlinux AUR](#)

On Arch you can build using the PKGBUILD provided in packaging directory or from the AUR package. To build manually, clone the repo and do:

Listing 1: Manual Install

```
rm -f dist/*  
/usr/bin/python -m build --wheel --no-isolation  
root_dest="/"   
./scripts/do-install $root_dest
```

When running as non-root then set root_dest a user writable directory

3.2 Dependencies

- Run Time :
 - python (3.13 or later)
 - passlib
 - bcrypt (aka python-bcrypt)
 - argon2-cffi (aka python-argon2-cffi)
 - cryptography (aka python-cryptography)

NB versions 1.1 and earlier used openssl - all newer version now use python passlib library.

- Building Package:
 - git
 - hatch (aka python-hatch)
 - wheel (aka python-wheel)
 - build (aka python-build)
 - installer (aka python-installer)
 - rsync
- Optional for building docs:

- sphinx
- texlive-latexextra (archlinux packaguing of texlive tools)

3.3 Philosophy

We follow the *live at head commit* philosophy as recommended by Google's Abseil team¹. This means we recommend using the latest commit on git master branch.

3.4 License

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¹ <https://abseil.io/about/philosophy#upgrade-support>

LICENSE

nginx_passwd provides a Basic Auth Password File Manager supporting new and legacy hash algorithms.

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